Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For

Every day you meet information all around you: how many learners came to school, how many goals a team scored, how much rain fell in a week, how many people voted. On its own, a jumbled list of such facts is hard to make sense of. Statistics is the method of collecting and organising information so that it can be understood and used.

Its purpose is not to gather numbers for their own sake. When information is collected and arranged neatly, it becomes easy to see what is happening, to compare one thing with another, and to make sensible decisions. A teacher collects the number of learners present each day; arranged in order, that information shows which day of the week attendance was best. A shopkeeper records what was sold; organised into a table, it shows which items are most wanted. This is what statistics is for: turning scattered information into something clear enough to read and act on.

So the heart of statistics is two steps: collecting the information, and organising it so that its meaning can be seen.

Suppose a class wants to know which colour of pencil is most common among its learners. First the information is collected: each learner names their pencil colour — blue, red, blue, green, red, blue, blue, green, red, blue. That list alone is hard to read, so the information is then organised.

Write each colour once, and count how many times it appears:

- Blue: 5
- Red: 3
- Green: 2

Now the meaning is clear at a glance: blue is the most common pencil colour. Statistics collected the information and organised it so a decision — which colour to buy more of — could be made easily.

Practice

1. Explain in your own words what statistics is, and state its two main steps.
2. A gardener notes the number of tomatoes picked from a plant each day: 4, 6, 4, 5, 6, 4, 5, 6. Organise this information by counting how many times each number appears.

Section 2: From Tallies to a Frequency Table

When a statistician wants to find out something — say, how many brothers and sisters your classmates have, or how many bottles were collected for recycling — the first thing to do is collect the information. Every single piece of information you gather is called a piece of data. But a long list of numbers is hard to read at a glance. To understand it, you need to organise it into a table, and the most useful table for this is a frequency table.

A frequency table records three things for each different value: the value itself, how you counted it using tally marks, and the frequency — the number of times that value appears.

Let's look at how tally marks work. Each time a value appears, you draw one short vertical stroke |. When you reach the fifth mark, you draw it across the four as a diagonal stroke, making a bundle of five: ||||. This bundle makes it much easier to count your marks at the end. So:

- four marks are |||| (four strokes)
- five marks are |||| (four strokes and one across)
- seven marks are |||| ||

- ten marks are |||| ||||

To build a frequency table, you go through your data one value at a time. For each value you meet, find its row in the table and add one tally mark. When you have finished the whole list, you count the tallies in each row and write that number in the frequency column. The tally and the frequency must always match, so counting carefully is the whole skill.

Let's follow one small collection of data all the way through. Ten students recorded how many pencils each had on their desk, and here is the list:

3, 1, 2, 3, 1, 3, 2, 1, 3, 2

There are ten students, so there are ten numbers. The values that appear are 1, 2 and 3. We set up a frequency table with a row for each of these values.

Start with a blank tally column. Now go through the list in order and add a mark each time a value appears.

- First number is 3: add one mark in row 3.
- Next is 1: add a mark in row 1.
- Next is 2: add a mark in row 2.
- Next is 3: add a mark in row 3.
- Next is 1: add a mark in row 1.
- Next is 3: add a mark in row 3.
- Next is 2: add a mark in row 2.
- Next is 1: add a mark in row 1.
- Next is 3: add a mark in row 3.
- Last is 2: add a mark in row 2.

Now count each row of tallies.

- Row 1: three marks, so frequency 3.
- Row 2: three marks, so frequency 3.
- Row 3: four marks, so frequency 4.

The completed frequency table looks like this:

Number of pencils	Tally	Frequency
1					3
2					3
3						4
Total		10

Check your work: the frequencies should add up to the total number of pieces of data. Here $3 + 3 + 4 = 10$, and there were indeed ten students. If they did not add up, you missed a mark or counted one wrongly — go back and find the mistake.

Here is a complete example for you to follow. Twelve students in a class were asked their shoe size, and the results were recorded as follows:

6, 7, 5, 6, 6, 7, 5, 6, 7, 5, 6, 7

Step 1 — Set up the table. The values that appear are 5, 6 and 7, so give each its own row.

Step 2 — Tally the data. Go through the list in order and add one mark per value.

- 6: one mark in row 6
- 7: one mark in row 7
- 5: one mark in row 5
- 6: one mark in row 6
- 6: one mark in row 6
- 7: one mark in row 7
- 5: one mark in row 5
- 6: one mark in row 6
- 7: one mark in row 7

- 5: one mark in row 5
- 6: one mark in row 6
- 7: one mark in row 7

Step 3 — Count the tallies.

- Row 5 has four marks, so frequency 4.
- Row 6 has five marks (a full bundle of five), so frequency 5.
- Row 7 has three marks, so frequency 3.

Step 4 — Check the total. 4 + 5 + 3 = 12, which matches the twelve students. The table is correct.

Shoe size	Tally	Frequency
5	||||	4
6	|||||	5
7	|||	3
Total		12

Once the table is complete you can read answers straight off it. The most common shoe size is 6, because it has the highest frequency (5). Five students wore size 6, four wore size 5, and three wore size 7.

Practice

1. Copy the tally marks below and write the frequency of each. Then find the total.

Colour	Tally
Red	|||
Blue	||||
Green	||

   What is the frequency of each colour, and how many pieces of data are there in total?
2. Prepare a frequency table for the following data, showing the value, the tally and the frequency:

   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

Check your work

Attempt every question before you read this page.

What Statistics Is For

1. Statistics is the method of collecting and organising information so that it can be understood and used. Its two main steps are collecting the information and organising it.
2. 4 appears 3 times, 5 appears 2 times, 6 appears 3 times.

   Count each number in the list: 4, 6, 4, 5, 6, 4, 5, 6. There are three 4s, two 5s and three 6s. Organising it this way shows that 4 and 6 were equally common, and 5 was least common.

From Tallies to a Frequency Table

1. Red: 3, Blue: 4 (a full bundle of five counts as 5, but here |||| is four strokes), Green: 2. Total = 3 + 4 + 2 = 9.

   Count the strokes in each row. Red has three strokes, so frequency 3. Blue has |||| which is four strokes, so frequency 4. Green has two strokes, so frequency 2. Add them: 3 + 4 + 2 = 9 pieces of data.
2. | Value | Tally | Frequency |
   |---|---|---|
   | 1 | \|\|\| | 3 |
   | 2 | \|\|\| | 3 |
   | 3 | \|\|\|\| | 4 |

Value	Tally	Frequency
4	|||	3
Total		13

List the values that appear: 1, 2, 3 and 4. Go through the list in order and tally each.

For 1: first 1, then 1 (fifth number), then 1 (twelfth number) — three 1s, frequency 3.
For 2: the seventh, eighth and thirteenth numbers are 2 — three 2s, frequency 3.
For 3: the third, sixth, ninth and tenth numbers are 3 — four 3s, frequency 4.
For 4: the fourth, fifth and eleventh numbers are 4 — three 4s, frequency 3.

Check the total: $3 + 3 + 4 + 3 = 13$, which matches the 13 numbers given.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-8dd7fbea	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____